

CURRICULUM VITAE

NYASHADZASHE NDIRIPO

Mechatronics Intern Applicant

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PERSONAL DETAILS

- Date of Birth: 19 September 2003
- National ID Number: 27-256033X27
- Nationality: Zimbabwean
- Marital Status: Single
- Driver's Licence: Class 4
- Availability: Currently available

CAREER OBJECTIVE

A City & Guilds electrical and electronics student with a strong personal interest in mechatronics, automation and control systems. Alongside my diploma studies, I have designed and built an independent sensor-actuated mechatronic module (a conveyor belt steering system using ultrasonic sensors and a linear actuator), programmed PLCs in ladder logic for real automated processes, and worked hands-on with electronics, circuits and CAD design. I am looking for an internship where I can bring this analytical, hands-on approach into real industrial automation and control projects, and keep learning from experienced engineers.

EDUCATION

- City & Guilds Level 5 Advanced Technician Diploma in Electrical and Electronics Engineering, currently in progress.
- City & Guilds Level 4 Diploma in Electrical and Electronics Engineering.
- City & Guilds Level 3 Diploma in Electrical and Electronics Engineering.
- City & Guilds Level 2 Certificate in Electrical and Electronics Engineering.
- Ordinary Level Certificate.

MECHATRONICS & AUTOMATION PROJECTS

- Designed an independent sensor-actuated mechatronic module, a conveyor belt steering system that uses ultrasonic sensors to detect belt drift and a linear actuator to self-correct it, modelled as a complete parametric CAD assembly built around real sourced components (bearings, actuator, sensors) with a PLC-based control system; learned how a mechatronic design moves from concept through to a build-ready engineering package.
- Designed, entered and tested a ladder logic PLC program in CODESYS V3.5 for an industrial mixing process, with two fill valves controlled by level sensors, a timed mixer ON/OFF cycle repeated 6 times using a counter, and then a timed drain valve, built using step bits, TON timers and a CTU counter; learned how to structure a full sequential automated process safely in ladder logic and verify it through online simulation.
- Programmed a PLC using ladder logic on SMT client software and SMT hardware, writing a basic garage door control program; learned the fundamentals of ladder logic and PLC-based automation.
- Wired a sequential power and control circuit to run four motors in sequence, and carried out control-circuit fault troubleshooting exercises; learned a structured approach to finding and fixing faults in automated control circuits.
- Built an ADC circuit on breadboard and then soldered the prototype onto a custom PCB; learned how a sensor interface circuit moves from a breadboard prototype to a finished, soldered board.
- Designed, simulated in Proteus and bench-tested analogue circuits, including an op-amp amplifier, an oscillator and an active filter; learned how theoretical circuit calculations compare against real simulated and measured behaviour.
- Designed and wired a star-delta forward/reverse motor control circuit, and carried out general motor maintenance tests; learned how motor starting and direction-control circuits work in practice.
- Diagnosed and repaired an electrical fault on a CNC milling machine in a corrective maintenance project; learned a structured fault-finding approach for industrial equipment.
- Microprocessor-Based Systems: designed and built a temperature-controlled fan system on breadboard using an Arduino, a DHT22 temperature/humidity sensor, an I2C 20x4 LCD display and a relay to switch the fan; learned how a microcontroller reads a sensor, displays live data and drives an output device as part of one working system.
- Wired a full house electrical installation for a six-roomed house on a wood panel, including radial and ring final circuits and DB wiring, then tested it through to working lights and plugs; learned how to wire and test a real installation to a professional standard.

WORK EXPERIENCE

Electrical Intern, Talspark Trading / Value Solar

2025 (1 year)

- Solar installation: fitted solar panels, inverters and battery banks for residential and small commercial clients, from mounting panels through to running and terminating DC and AC cabling.
- Wired and connected inverters, charge controllers and battery banks, then tested completed systems to confirm correct polarity, voltage and load performance before handover.
- Carried out site call-outs on completed solar installations to trace and fix faults, and re-tested systems after repair to confirm they were working correctly.
- Industrial and general electrical work: wired panels and control circuits for machines and equipment, and carried out light fittings, conduit/tubing, and general wiring work on site.
- Rewired distribution boards (DB rewiring), including replacing and re-terminating circuits, and diagnosed electrical faults across both solar and general installations.
- Worked closely with the operations manager and senior technicians on more complex jobs, and worked independently on routine wiring, testing and fault-finding tasks.

SOFTWARE & TOOLS

Autodesk Fusion 360, FreeCAD, Proteus, KiCad, AutoCAD Electrical, CODESYS, LogixPro 500, XG5000.

SKILLS & COMPETENCIES

- PLC ladder logic programming (CODESYS, LogixPro, XG5000) and automated control system design.
- Mechatronic design: sensors, actuators and parametric CAD (FreeCAD, Fusion 360, AutoCAD Electrical).
- Circuit design and simulation (Proteus, KiCad), digital electronics and ADC/PCB prototyping.
- Motor control circuits (star-delta, forward/reverse, sequential control) and general motor maintenance.
- Electrical wiring, fault-finding and repair for domestic, solar and industrial installations.
- Microcontroller (Arduino) work and basic C programming.
- Strong analytical and structured troubleshooting approach; quick to learn new systems and equipment.
- Works well under pressure with minimal supervision; clear communication with supervisors and team members.

REFERENCES

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Simbarashe Jakopo, Operations Manager, Talspark Trading / Value Solar | Tel: +263 773 261 524 | Email: jakoposimba01@gmail.com